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RV COLLEGE OF ENGINEERING®
 (An Autonomous Institution affiliated to VTU)
 VII Semester B. E. Examinations Feb/Mar-2023

Computer Science and Engineering
COMPUTER GRAPHICS AND VIRTUAL REALITY

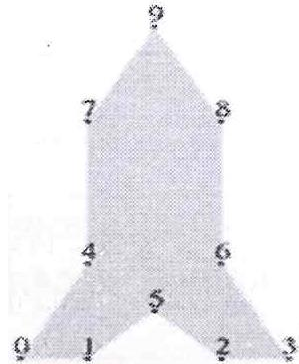
Time: 03 Hours

Maximum Marks: 100

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	Describe the Three I's of Virtual Reality.	02
	1.2	Consider the object/polygon to be drawn: Which calling sequence of these vertices (using <i>glVertex2fv</i>) results in the shape (given on the right) if OpenGL function <i>glBegin(GL_TRIANGLE_FAN)</i> and <i>glEnd()</i> ? Write the OpenGL Code Segment. Refer Fig. 12	
			
		Fig. 12	02
1.3		Derive the general form of the matrix for rotation of an object about a fixed point $P(h,k)$ obtained from composite transformations.	02
1.4		Give the code segment in OpenGL to construct a triangle with the fill color as green and the boundary red colored.	02
1.5		Differentiate between different types of Sound Displays.	02
1.6		Draw the viewing volume for Orthographic and Perspective projections with associated OpenGL functions.	02
1.7		A cube is defined by 8 vertices: $A(0,4,0)$, $B(0,4,4)$, $C(4,4,0)$, $D(4,4,4)$, $E(4,0,0)$, $F(0,0,4)$, $G(4,0,4)$, $O(0,0,0)$. Apply translation on the same where the given translation distances are $tx = 2$, $ty = 4$, $tz = 6$. Draw the resultant cube after applying the transformation.	02
1.8		Write the neat sketches to represent the steps involved in 3-D rotation of an object about an axis that is not parallel to one of the coordinate axes.	02
1.9		Differentiate between the two major classification of Visible surface detection methods.	02

1.10	If the Scale Factors $S_x \neq S_y$ then it is known as _____ Scaling and $S_x = S_y$ then it is _____ Scaling respectively.	01
1.11	_____ line clipping algorithm uses parametric representation of line segments and _____ line clipping algorithm uses region code approach respectively.	01

PART-B

2	a	Write a neat diagram indicating the library organization of the OpenGL interface Explain the functionality provided by each of the libraries bringing out the main features.	06
	b	Write a OpenGL program segment to construct 2D Sierpinski Gasket by Recursive approach. Explain how we can extend this idea to construct 3D Sierpinski Gasket.	06
	c	Discuss with a neat diagram the five classic components of a VR system.	04
3	a	Develop an OpenGL program to create the following hierarchical menu and attach the corresponding operations to each menu with mouse left button. Write the complete program with appropriate code for each Menu and Submenu entries. Refer Fig. 3b <pre> graph LR subgraph MenuBox [] direction TB Object[Object] Color[Color] Quit[Quit] end subgraph TargetBox1 [] direction TB Sphere[3D Sphere (Approximation)] end subgraph TargetBox2 [] direction TB Red[Red] Green[Green] Blue[Blue] end Object --> Sphere Color --> Red Color --> Green Color --> Blue </pre> Fig. 3b	10
	b	All 3D trackers, regardless of the technology they use, share a number of very important performance parameters. Define these performance parameters and illustrate the same with neat graphical representations.	06
OR			
4	a	Write OpenGL code segment to model a car like figure using display lists and move a car from one end of the screen to another end. Provide a keyboard interface to move and control the speed of the car.	06
	b	Compare the different modes by which input devices provide input to an application program with schematic diagrams.	05
	c	Discuss the features of Haptic Feedback and differentiate between the two modalities.	05

5 a b	Design and develop an Interactive Computer Graphics program using appropriate OpenGL transformation functions to perform the following operations (independently) on a tetrahedron with minimal OpenGL calls: Simulate the following operations independently for the object (indicate the coordinate axis): - Draw the Tetrahedron with the coordinate axis. - Simulate orientation of the tetrahedron with left mouse button (left button). - Simulate translation (provide keyboard interface --- left and right arrow keys) - Simulate the Scaling operation zoom in and out with a third button (middle button) Examine the working principle of Sutherland-Hodgeman polygon clipping algorithm for the polygon given in Fig. 5b. against the clipping window. Hand-simulate all the intermediate processing steps and indicate the same with neat sketches. Fig. 5b OR	10 06
6 a b	Apply the Liang-Barsky algorithm to clip the lines AB from $(11,6)$ to $(11,10)$, CD from $(3,7)$ to $(3,10)$ and EF from $(2,3)$ to $(8,4)$ against the rectangular window whose lower left hand corner $L(1,2)$ and upper right hand corner $R(9,8)$. Indicate the results graphically before and after clipping. Examine how pixel spans could be filled using Boundary-fill approach (for the 4 – <i>connected</i> fill region) for the following polygon in Fig. 6b. Illustrate the filled pixel spans of the polygon pictorially for 4-connected filled region. Indicate the complete trace of the same using appropriate data structure. Fig. 6b	06 05

	c	Write a C program segment to draw a line between the given endpoints using DDA algorithm.	05
7	a	The triangle $[ABC] = [(5, 3, 7), (4, 2, 8), (10, 4, 9)]$ is required to be magnified to four times of its original size while keeping $C(10, 4, 9)$ fixed. Write the C-program to perform the operation without using in-built OpenGL Functions.	06
	b	Write the complete taxonomy of Classification of Planar Geometric projections. Explain the characteristic features of each type with the aid of neat sketches.	05
	c	Illustrate the working principle of Depth Buffer algorithm to remove hidden surfaces with aid of neat sketch and pseudo code.	05
8	a	Discuss the features of Geometric modeling, illustrate in detail the various aspects involved. Summarize several methods by which Object surfaces can be constructed.	06
	b	Discuss the different components of VR Modeling Cycle with the aid of a neat diagram.	05
	c	“Object hierarchies define groups of objects which move together as a whole, but whose parts can also move independently”. Validate this statement with the help of an example.	05